

Name: _____

TRANSFORMATIONS, (TRANSLATION)

Date: _____

LO: I can translate shapes using a column vector.

Translation

A **translation** slides a shape without rotating or flipping it. It is described by a **column vector** that shows the horizontal and vertical movement.

Column vector: The top number is the horizontal shift (positive = right) and the bottom number is the vertical shift (positive = up).

Quick examples

●	Vector (3, 2) means 3 right and 2 up	positive values
●	Vector (-2, -4) means 2 left and 4 down	negative values
●	(5, 1) translated by (3, -2) gives (8, -1)	add the vector
●	Vector (3, 2) means 3 up and 2 right	top is horizontal, bottom is vertical

Key idea

Add the column vector to each vertex: new $x = x +$ horizontal, new $y = y +$ vertical.

$$(x, y) + (a, b) = (x+a, y+b)$$

Every point moves by the same vector

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - positive translation

Translate (2, 3) by vector (4, 1).

Add horizontal: $2 + 4 = 6$

Add vertical: $3 + 1 = 4$

Image: (6, 4)

(6, 4)

Simply add the vector components to the coordinates.

Example 2 - negative components

Translate (5, 7) by vector (-3, -2).

Horizontal: $5 + (-3) = 2$

Vertical: $7 + (-2) = 5$

Image: (2, 5)

(2, 5)

Negative values mean left and down.

Example 3 - finding the vector

A is at (1, 4) and its image A' is at (5, 2). Find the translation vector.

Horizontal: $5 - 1 = 4$

Vertical: $2 - 4 = -2$

Vector = (4, -2)

(4, -2)

Subtract original from image to find the vector.

Example 4 - translating a shape

Translate triangle (1,1), (3,1), (2,4) by vector (-2, 3).

$(1,1) + (-2,3) = (-1, 4)$

$(3,1) + (-2,3) = (1, 4)$

$(2,4) + (-2,3) = (0, 7)$

(-1,4), (1,4), (0,7)

Apply the same vector to every vertex.

YOUR TURN – PRACTICE (deliberate practice)

1. Positive translations

Translate each point by the given vector.

- a) (1, 2) by (3, 4)
- b) (0, 5) by (2, 1)
- c) (3, 0) by (5, 3)
- d) (4, 4) by (1, 2)

2. Negative translations

Translate each point.

- a) (5, 3) by (-2, -1)
- b) (4, 6) by (-4, -3)
- c) (7, 2) by (-3, 4)
- d) (1, 8) by (2, -5)

3. Finding the vector

Find the translation vector from A to B.

- a) A(2, 3) to B(5, 7)
- b) A(6, 4) to B(2, 1)
- c) A(-1, 3) to B(4, -2)
- d) A(0, 0) to B(-3, 5)

4. Translating shapes

Translate the triangle by the vector.

- a) (0,0), (2,0), (1,3) by (4, 1)
- b) (1,1), (4,1), (4,3) by (-2, 3)
- c) (3,2), (5,2), (5,5) by (-1, -2)
- d) (2,4), (6,4), (4,7) by (1, -4)

5. Mixed translation problems

Solve each problem.

- a) (3, 5) by (-1, -3) b) Image is (7, 2), vector was (4, -3). c) Translate (0, 0) by (a, b) then by (-d) (2, 1) translated to (5, 4). Then (5, e) (4, -2) by (3, 5) then by (-1, -2).
 Original? a, -b). Final position? 4) to (8, 7). Combined vector? Final?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Translation by (3, -2) means 3 right and 2 down ☐

Correct answer: _____

Reason: _____

b) The top of the vector is the vertical shift ☐

Correct answer: _____

Reason: _____

c) Translation preserves shape, size, and orientation ☐

Correct answer: _____

Reason: _____

d) Translating by (0, 0) moves the shape to the origin ☐

Correct answer: _____

Reason: _____

e) Vector from A to B = B coordinates minus A coordinates ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Shape S is translated by vector (3, -2) to give S', then S' is translated by vector (-5, 4) to give S''. Find the single vector that maps S to S''. If a vertex of S is at (2, 6), find its final position.

Expression: _____

Simplified: _____

b) A square has vertices A(1,1), B(4,1), C(4,4), D(1,4). After a translation, A maps to (3, -1). Find the translation vector and the coordinates of B', C', and D'.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
 Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

